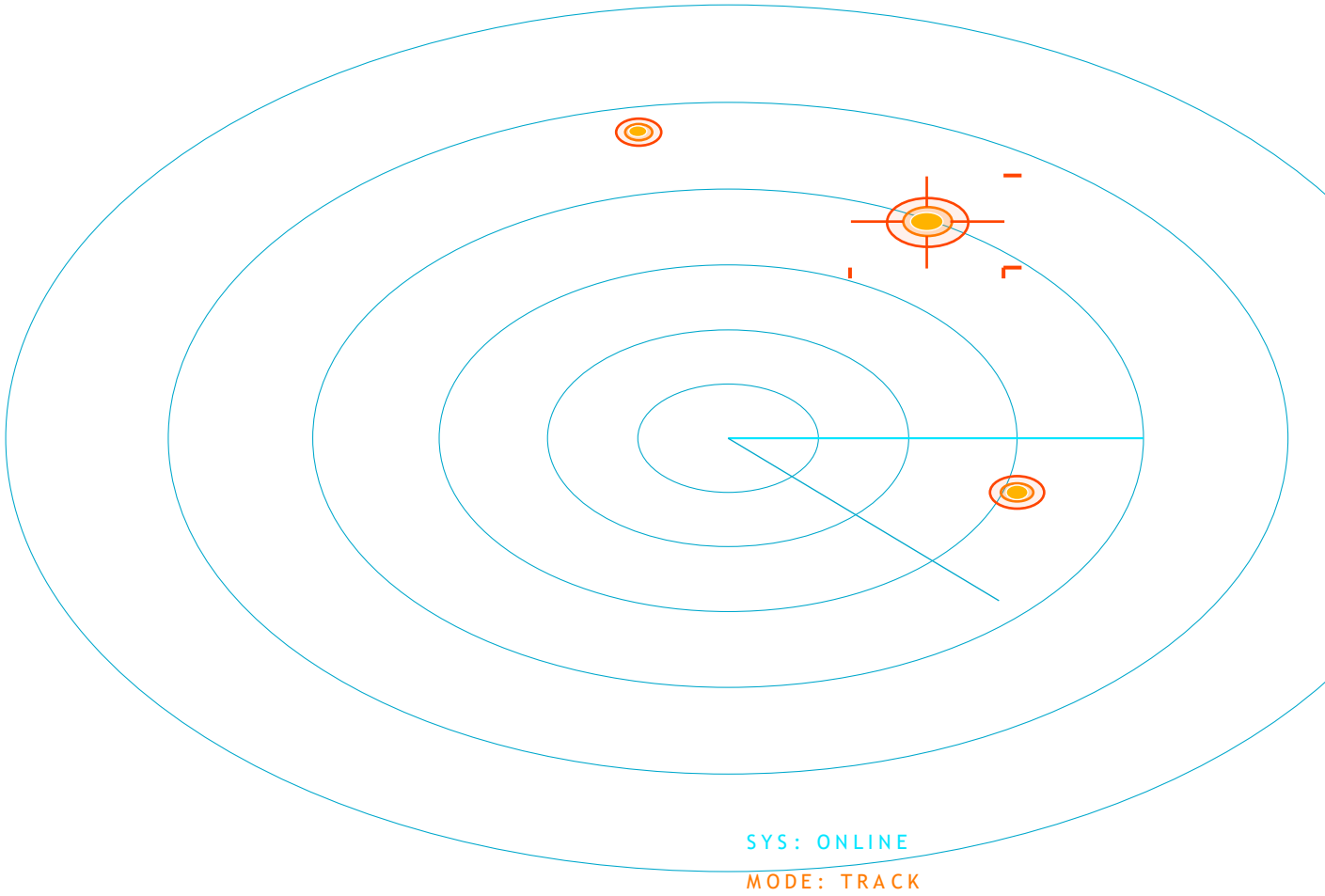
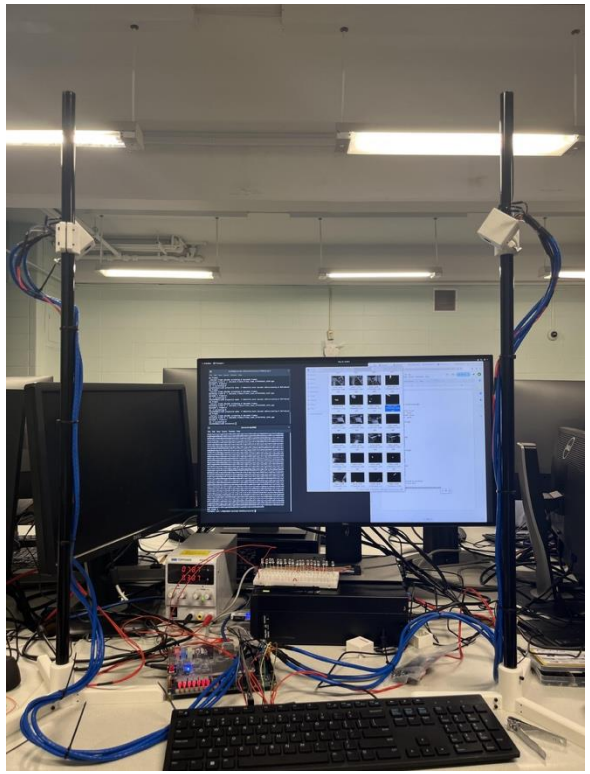
UNCALIBRATED STEREO MOTION TRACKING

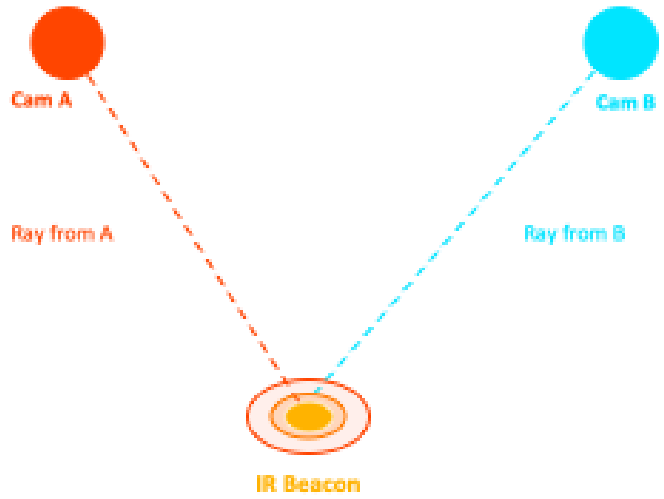


System Overview

- Image processing on FPGA: OV7670 data capture, pixel thresholding and IR beacon centroid detection
- Uncalibrated stereo in software: pre-calibrated camera intrinsics; everything else found dynamically.
- Functionality: Realtime 30 FPS IR beacon tracking with mm-scale precision

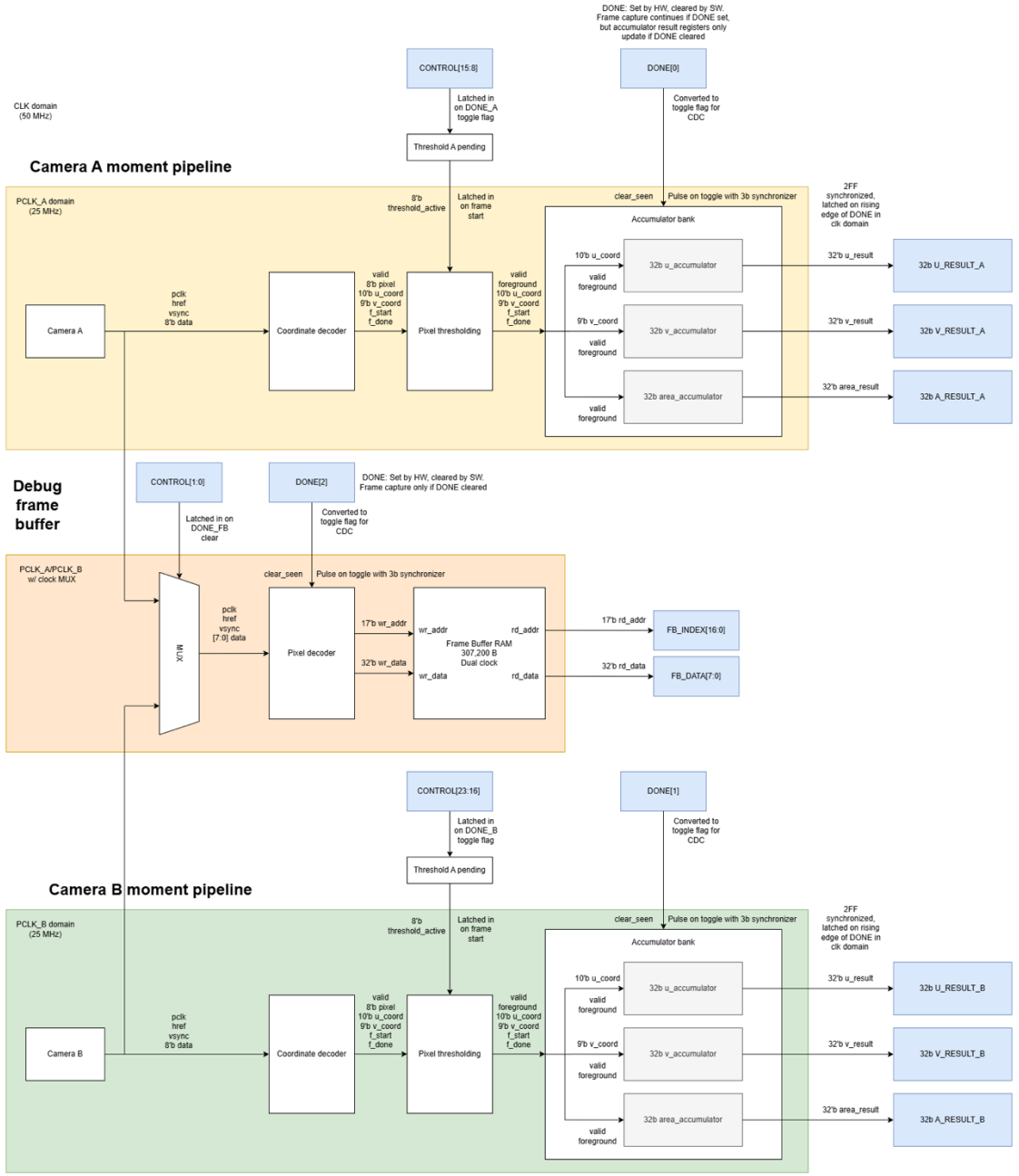




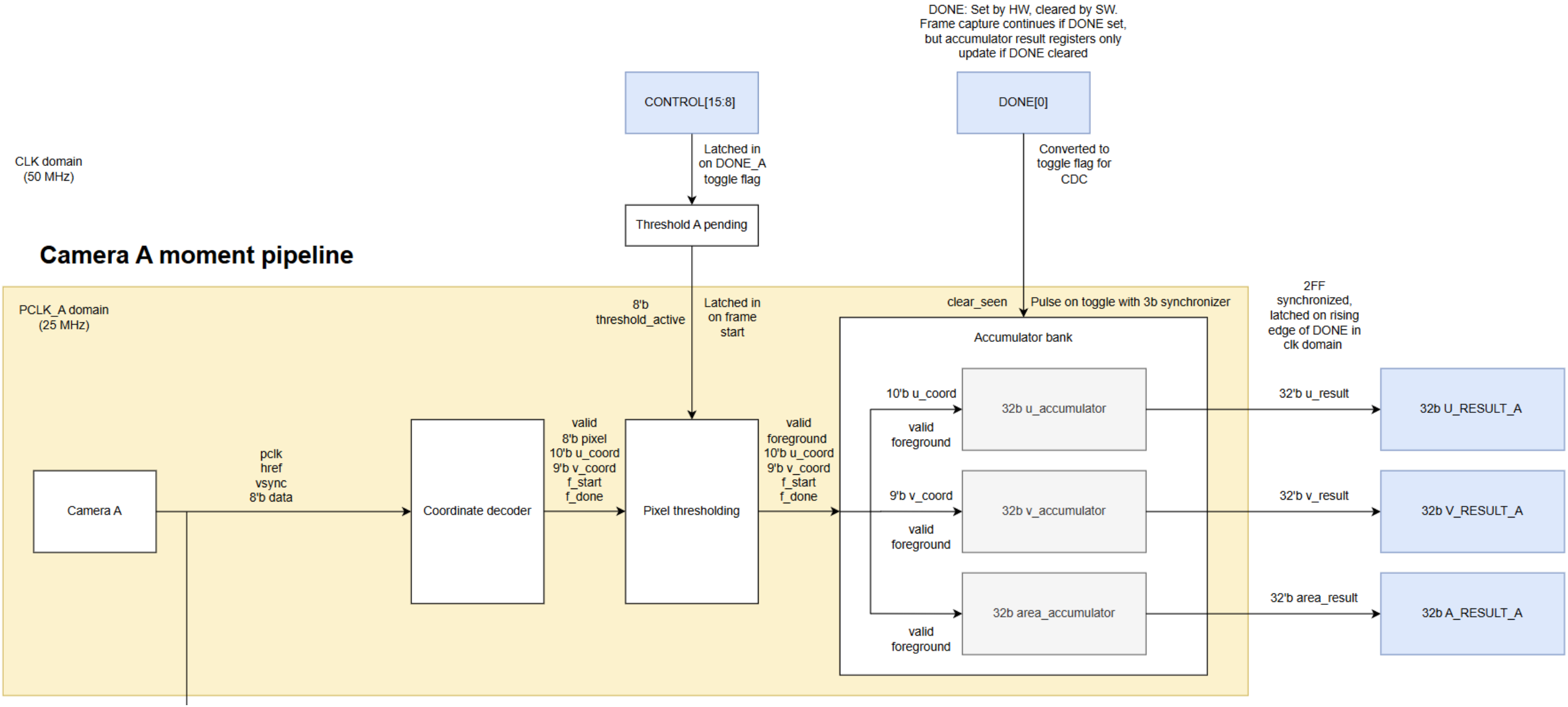


Hardware

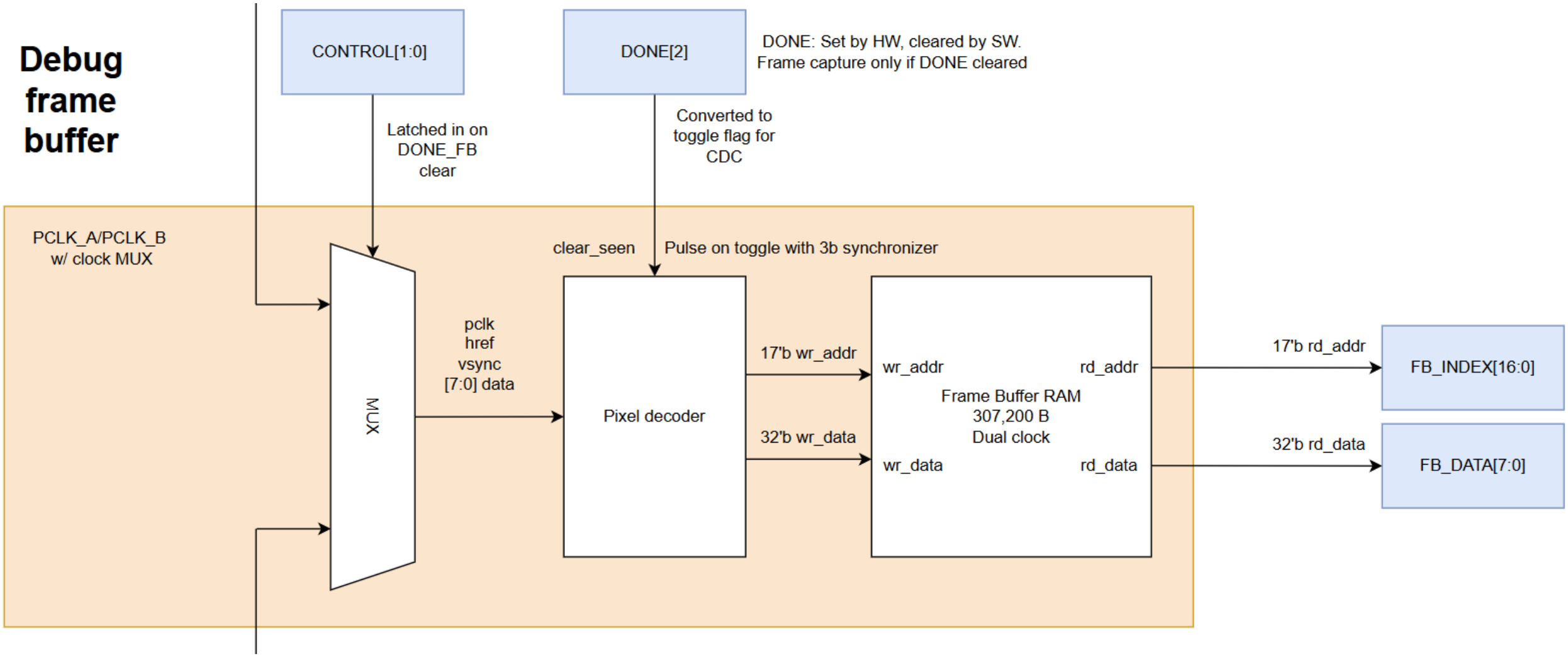
- Two moment accumulation pipelines and a debug frame buffer for photo capture



Camera A moment pipeline



Debug frame buffer



Pinout

Camera A:

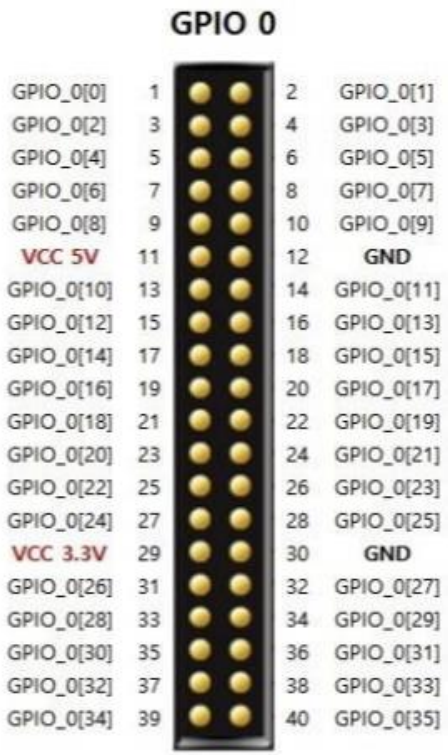
- GPIO_0[7:0] : DATA
- GPIO_0[8] : HREF
- GPIO_0[9]: VSYNC
- GPIO_0[10]: PCLK
- GPIO_0[26]: XCLK
- GPIO_0[35:27]: Software PIO
- GPIO_0[22:25]: Software I2C
- 22 = SCL
- 23 = SDA

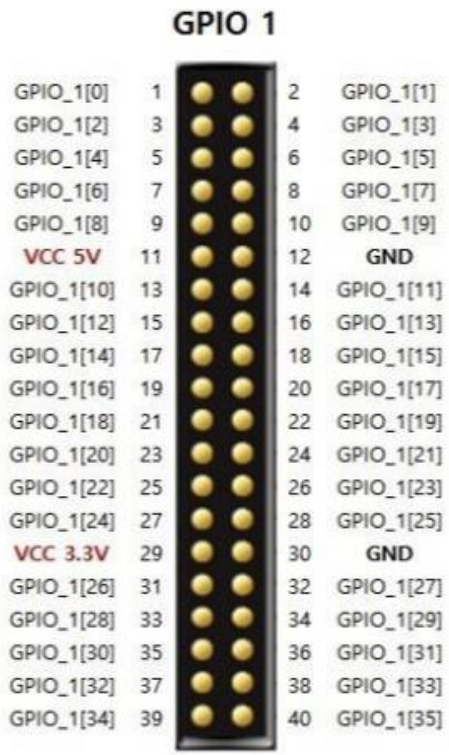
Camera B:

- Mirrored on GPIO_1

LEDR:

- [9]: Unused
- [8]: Always on unless reset
- [7:6]: Frame capture select mode (00 = OFF, 01 = A, 10 = B)
- [5:3]: STATUS (whether camera in capture state)
- [2:0]: DONE (frame_done, b_done, a_done)

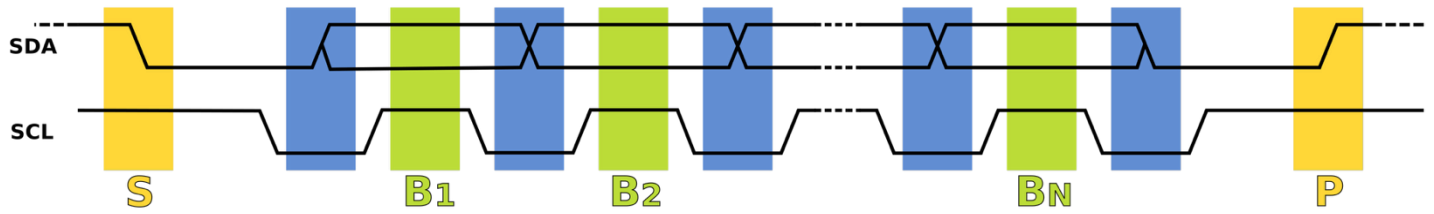


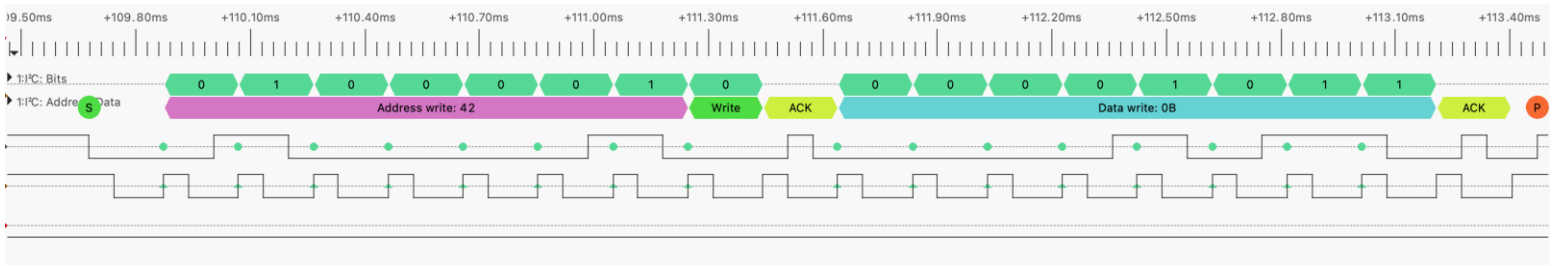


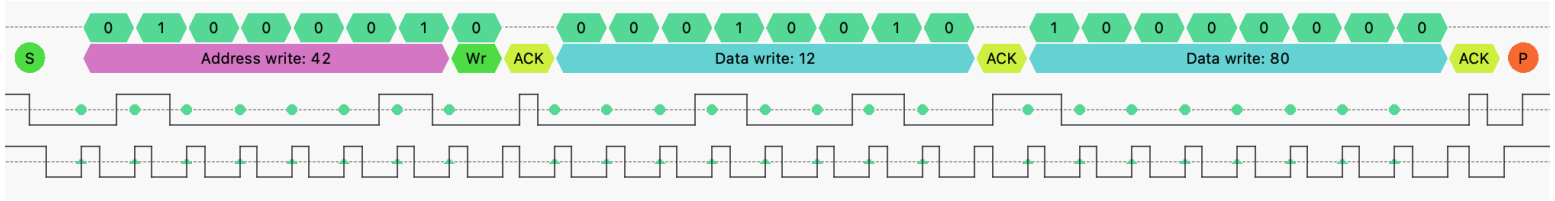
Add small description of custom cables

Camera Control

- Software bit-banged I2C using HPS-controlled PIO
- Device address 0x21
- R/W to control registers



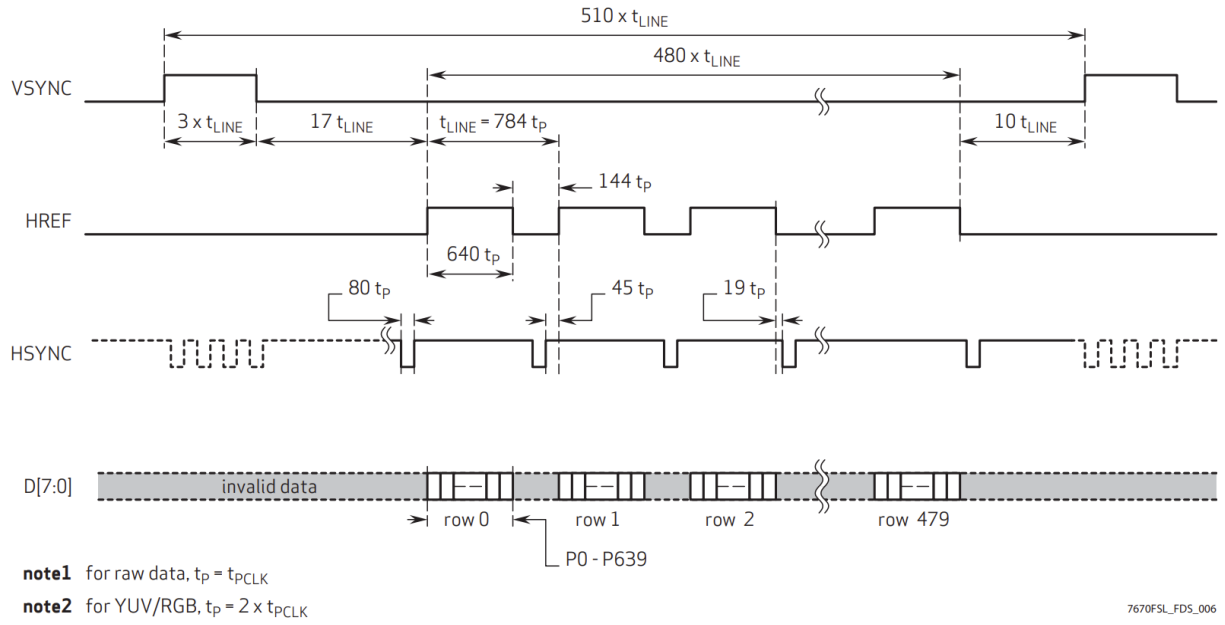


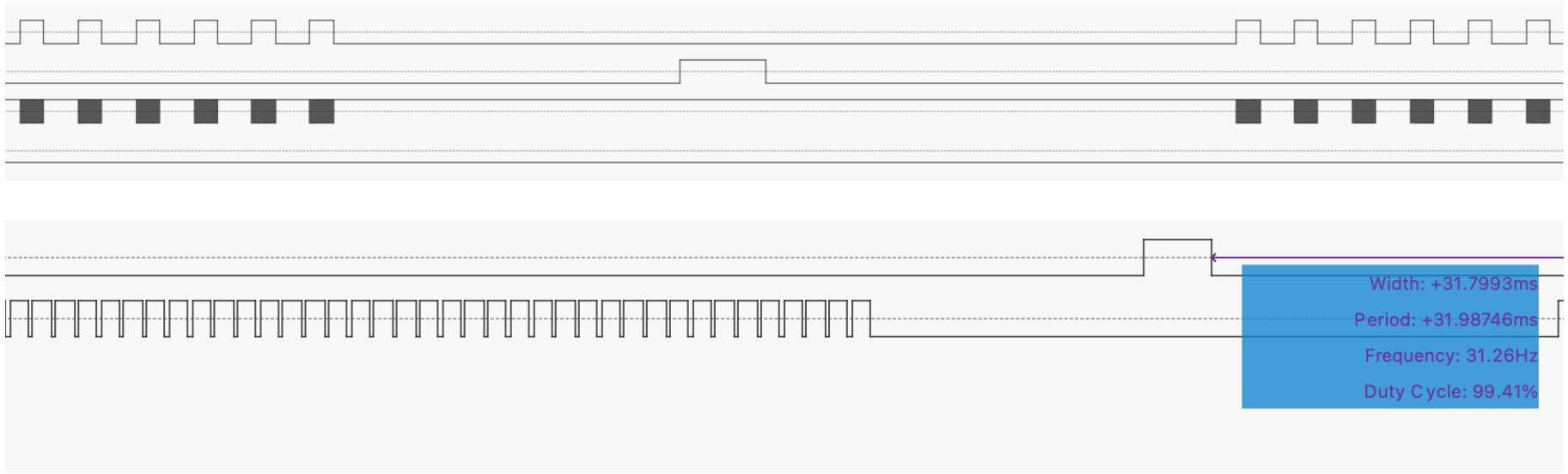


VGA Timing

- YUYV format with full VGA
- 31.26 FPS achieved with PCLK = 25 MHz

Figure 6 VGA Frame Timing





Hardware/Software Interface (1/2)

- Screenshot platform designer

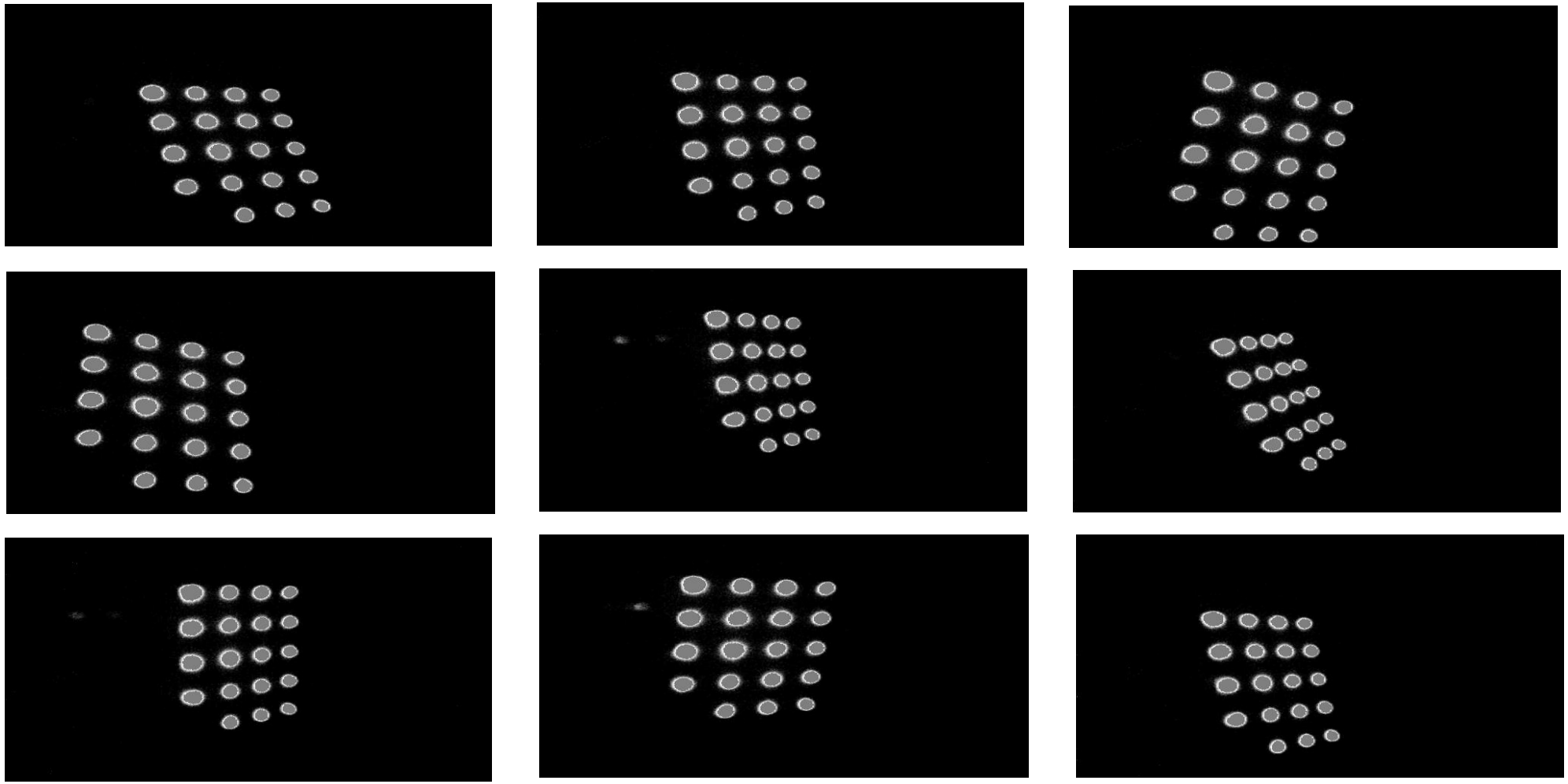
Hardware/Software Interface (2/2)

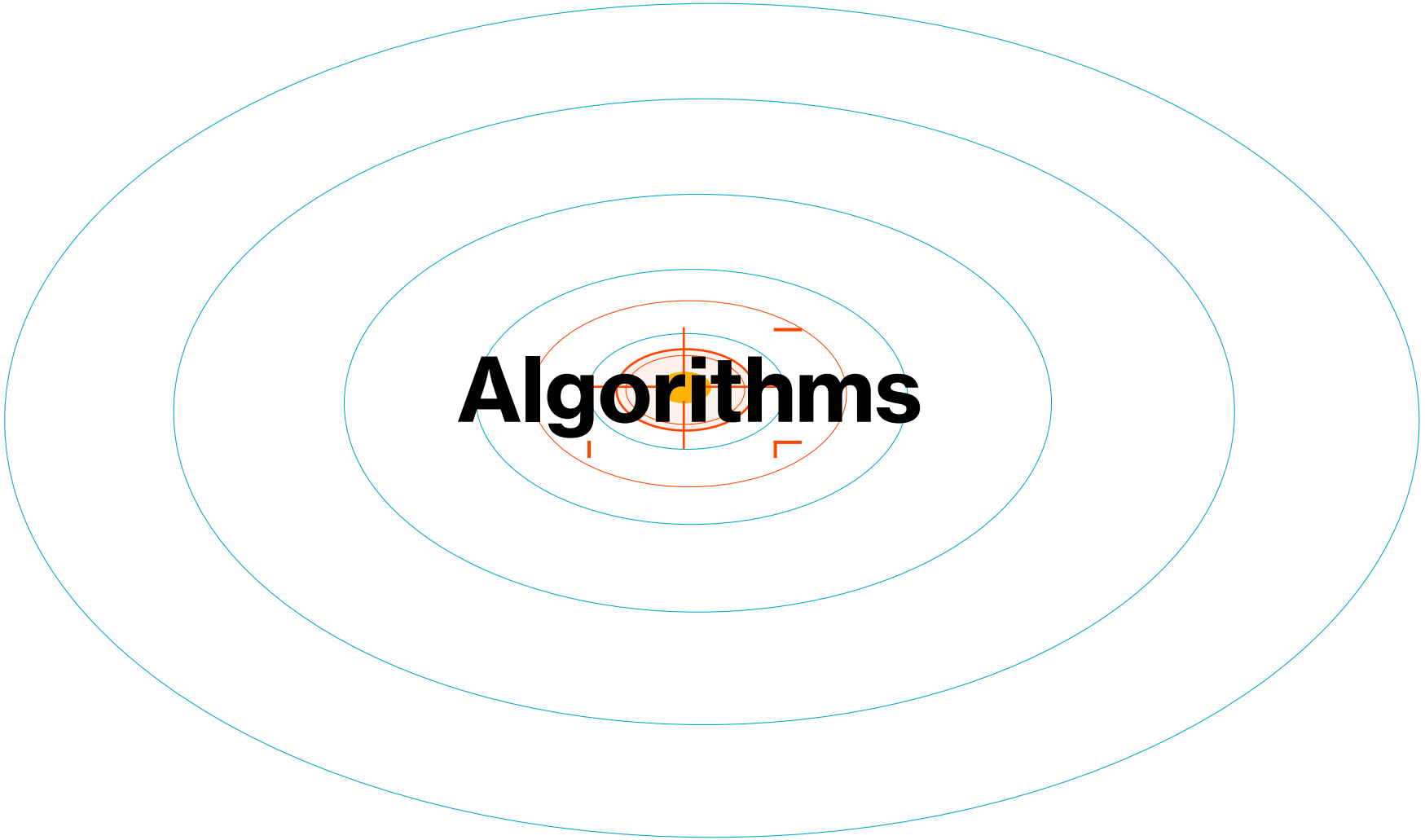
Register Map

Address	Register	Access	Meaning
0	AREA_A	R	Camera A foreground area
1	U_A	R	Camera A foreground u/x moment sum
2	V_A	R	Camera A foreground v/y moment sum
3	AREA_B	R	Camera B foreground area
4	U_B	R	Camera B foreground u/x moment sum
5	V_B	R	Camera B foreground v/y moment sum
6	DONE	R/W	bit 0: A, bit 1: B, bit 2: frame buffer
7	CONTROL	R/W	[1:0] frame source: 0 none, 1 A, 2 B; [15:8] threshold A, [23:16] threshold B
8	FB_INDEX	R/W	Frame buffer word index, 0 .. 76799
9	FB_DATA	R	Four pixels packed as {p3, p2, p1, p0}

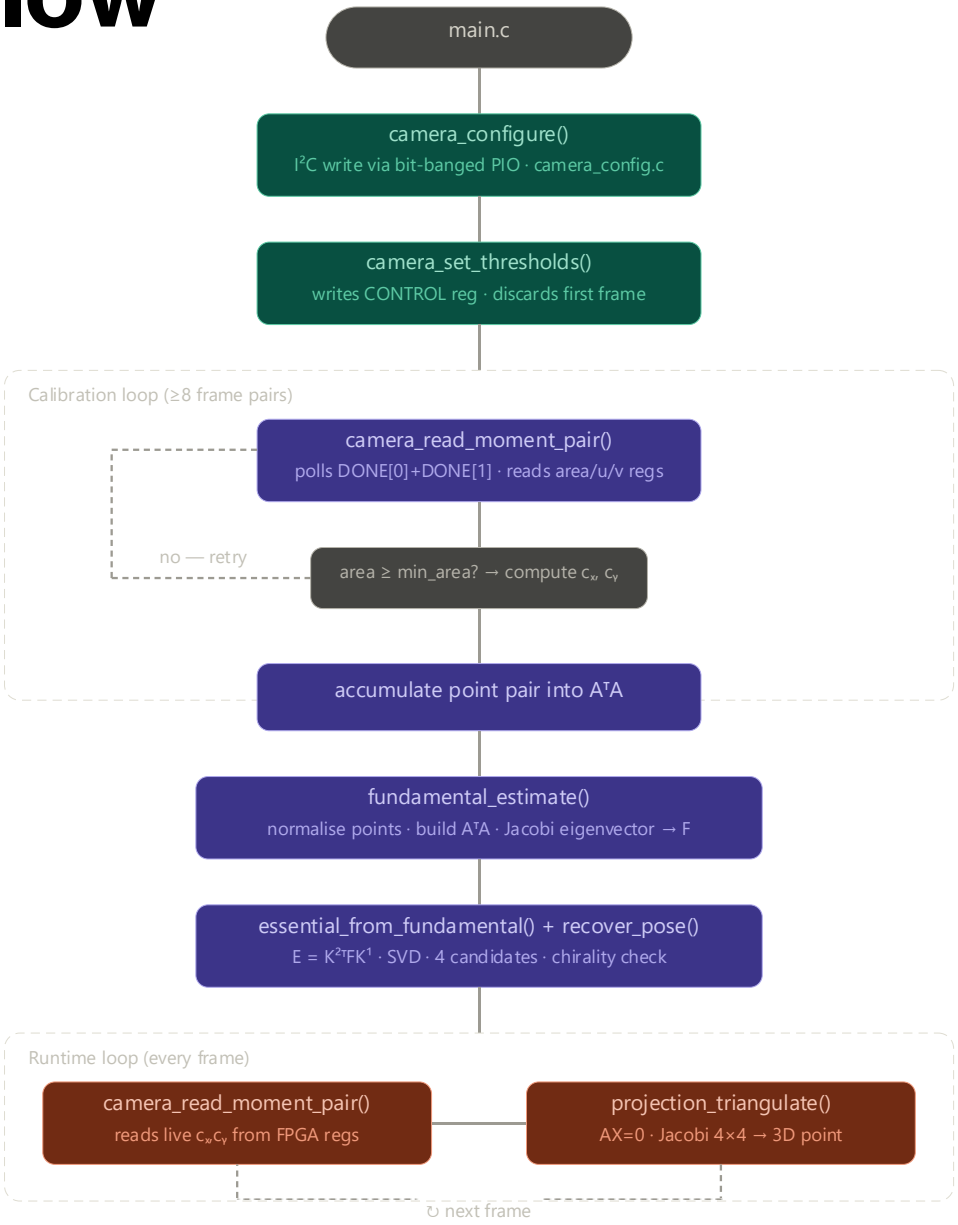
Test captures:

- Frame buffer captures from both OV7670 cameras via debug interface




Algorithms

Code Control Flow



Camera calibration

$$u = f_x \frac{x_c}{z_c} + o_x \qquad v = f_y \frac{y_c}{z_c} + o_y$$

Homogenous coordinates of (u, v):

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \equiv \begin{bmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{bmatrix} \equiv \begin{bmatrix} z_c u \\ z_c v \\ z_c \end{bmatrix} = \begin{bmatrix} f_x x_c + z_c o_x \\ f_y y_c + z_c o_y \\ z_c \end{bmatrix} = \begin{bmatrix} f_x & 0 & o_x & 0 \\ 0 & f_y & o_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix}$$

where: $(u, v) = (\tilde{u}/\tilde{w}, \tilde{v}/\tilde{w})$

Image Coordinates

$$\mathbf{u} = \begin{bmatrix} u \\ v \end{bmatrix}$$

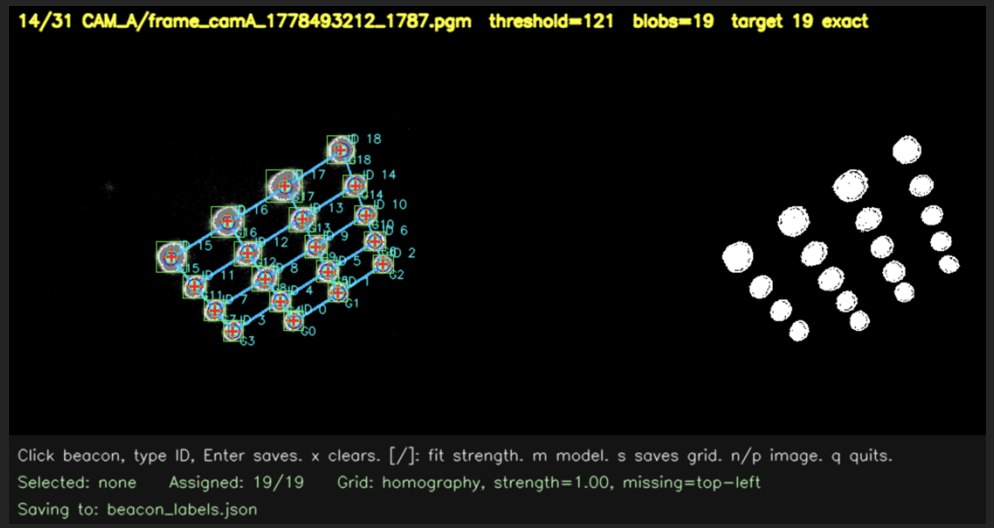
M_{int}

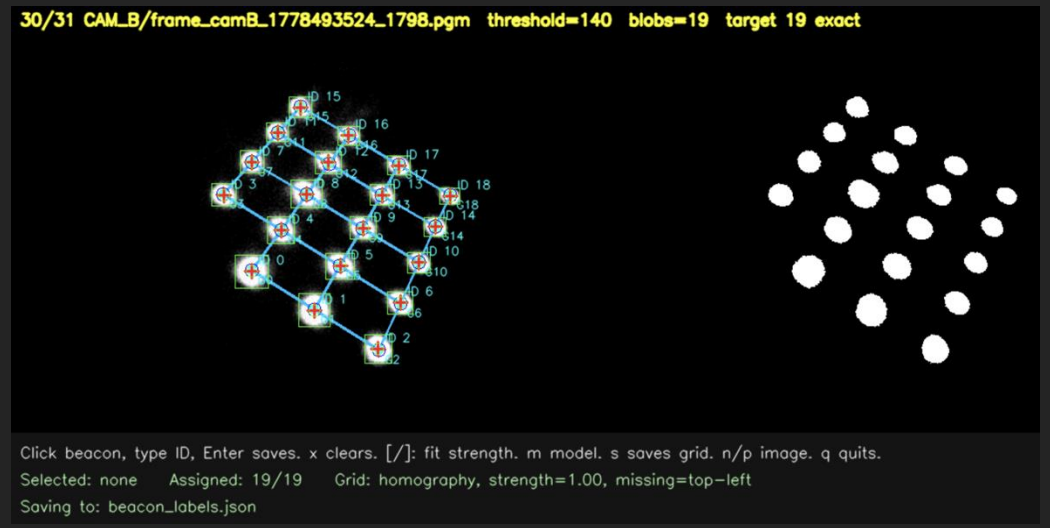
Perspective Projection

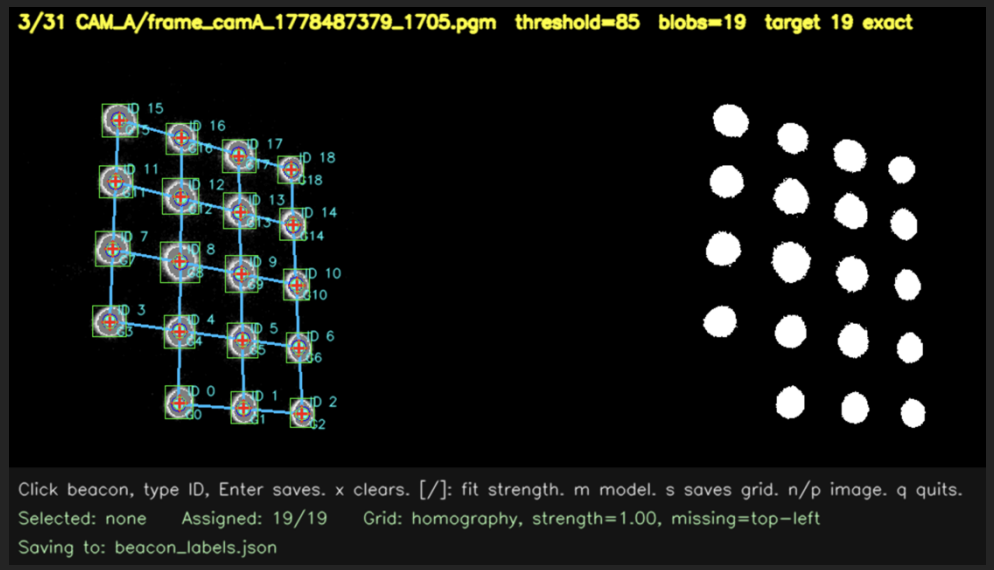
Camera Coordinates

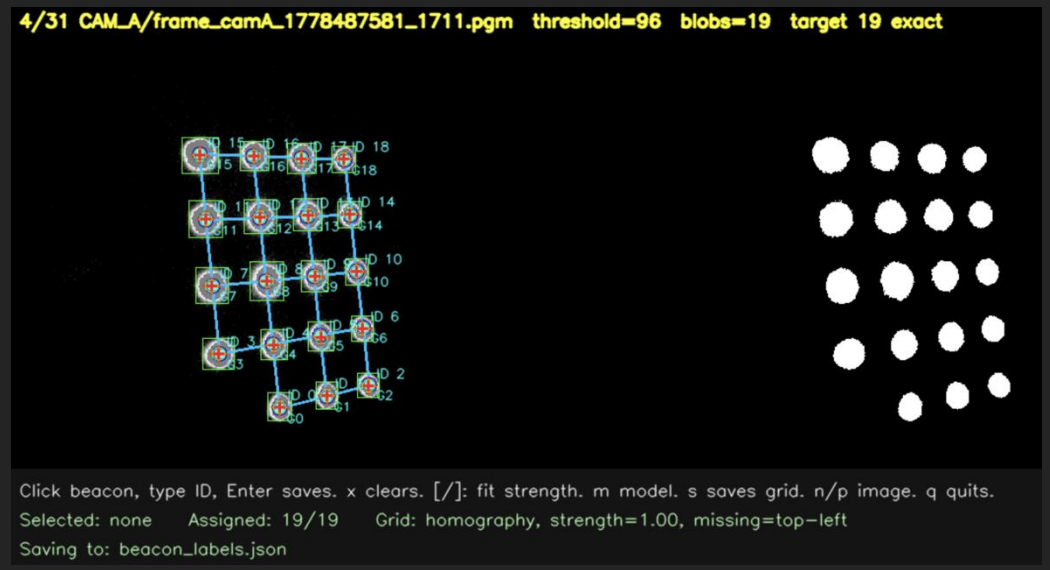
$$\mathbf{x}_c = \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix}$$

Camera calibration

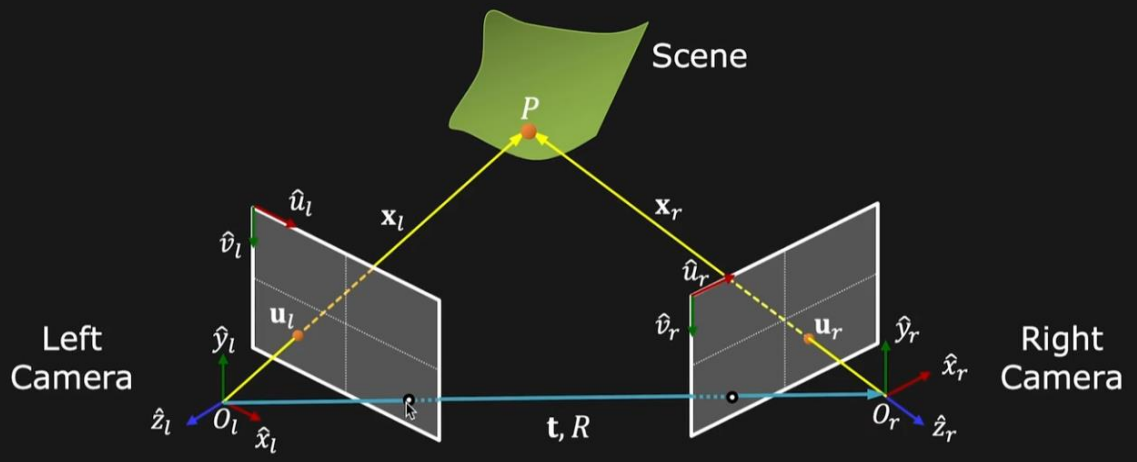


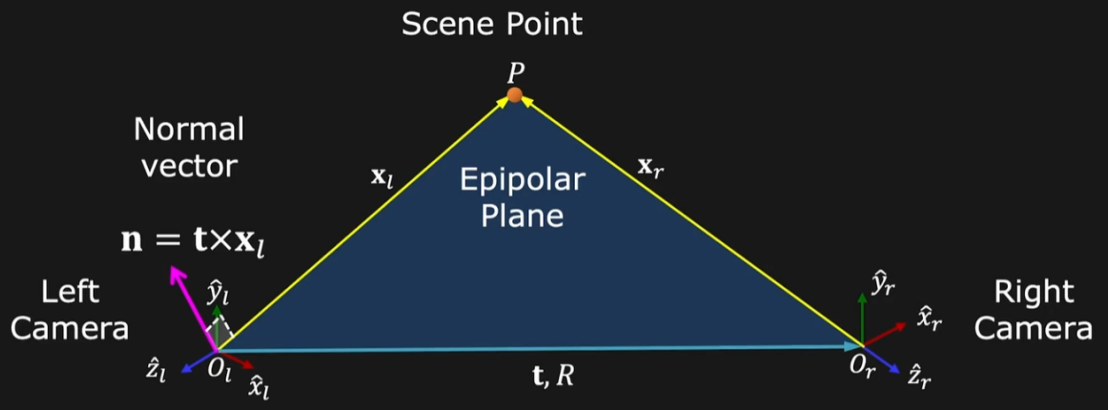






Epipolar geometry





Vector normal to the epipolar plane: $\mathbf{n} = \mathbf{t} \times \mathbf{x}_l$

Dot product of $\mathbf{n}$ and $\mathbf{x}_l$ (perpendicular vectors) is zero:

$$\mathbf{x}_l \cdot (\mathbf{t} \times \mathbf{x}_l) = 0$$

$$\begin{bmatrix} x_l & y_l & z_l \end{bmatrix} \left(\begin{bmatrix} 0 & -t_z & t_y \\ t_z & 0 & -t_x \\ -t_y & t_x & 0 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} + \begin{bmatrix} 0 & -t_z & t_y \\ t_z & 0 & -t_x \\ -t_y & t_x & 0 \end{bmatrix} \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} \right) = 0$$

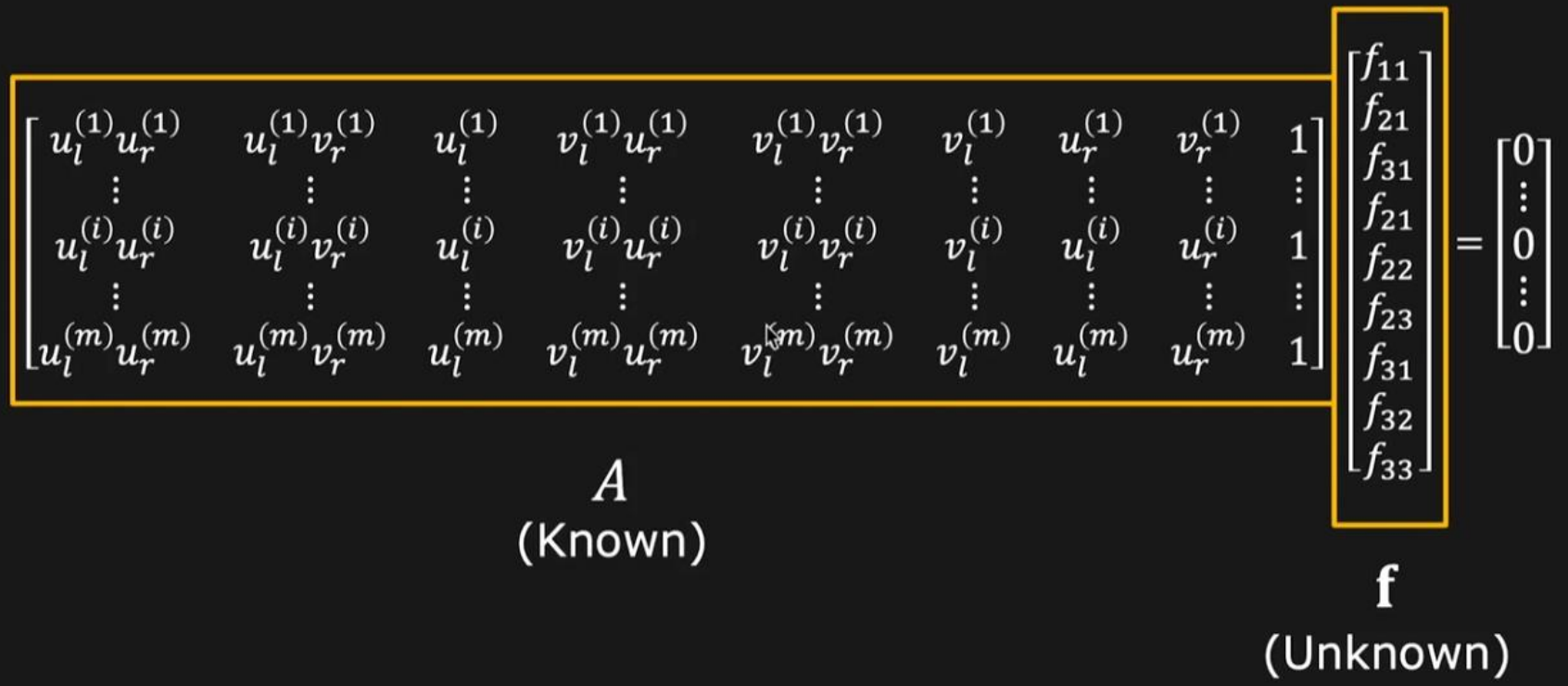
Essential and Fundamental Matrices

$$\begin{bmatrix} x_l & y_l & z_l \end{bmatrix} \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = 0$$

$$\begin{bmatrix} u_l & v_l & 1 \end{bmatrix} \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} = 0$$

Fundamental Matrix F

Fundamental Matrix Estimation



$$\min_{\mathbf{f}} \|A\mathbf{f}\|^2 \quad \text{such that } \|\mathbf{f}\|^2 = 1$$

Jacobi Least-Squares Solver

Goal: Estimate the fundamental matrix F from stereo correspondences.

Epipolar constraint:

$$x_2^T F x_1 = 0$$

Each matched point pair contributes one row:

$$[x_2x_1,\ x_2y_1,\ x_2,\ y_2x_1,\ y_2y_1,\ y_2,\ x_1,\ y_1,\ 1]$$

All rows form the homogeneous system:

$$Af \approx 0$$

Convert into eigenvalue problem:

$$A^T A f = \lambda f$$

The desired solution is the eigenvector corresponding to the smallest eigenvalue.

Jacobi Least-Squares Solver

1. Construct symmetric matrix:

$$A^T A$$

2. Find largest off-diagonal element:

$$a_{pq}$$

3. Compute Jacobi rotation:

$$\tau = \frac{a_{qq} - a_{pp}}{2a_{pq}}$$

$$t = \frac{\text{sign}(\tau)}{|\tau| + \sqrt{1 + \tau^2}}$$

4. Rotate rows and columns to eliminate a_{pq}
5. Repeat until matrix becomes diagonal
6. Extract eigenvector with smallest eigenvalue
7. Reshape into the 3×3 fundamental matrix F

Jacobi SVD

The estimated fundamental matrix is converted into an essential matrix:

$$E = K_2^T F K_1$$

A valid essential matrix must satisfy:

$$E = U \Sigma V^T$$

with constrained singular values:

$$\Sigma = \text{diag}(s, s, 0)$$

The algorithm computes:

$$E^T E$$

and applies Jacobi diagonalization again.

Eigenvalues correspond to:

$$\sigma_i^2$$

while eigenvectors form:

$$V$$

Singular values are recovered through:

$$\sigma_i = \sqrt{\lambda_i}$$

and left singular vectors are computed using:

$$U_i = \frac{EV_i}{\sigma_i}$$

Pose Recovery

1. Compute candidate rotations:

$$R_1 = UWV^T$$

$$R_2 = UW^TV^T$$

2. Extract translation vector:

$$t = U[:, 2]$$

3. Generate four possible camera poses:

$$(R_1, +t),\ (R_1, -t),\ (R_2, +t),\ (R_2, -t)$$

4. Perform chirality check using triangulated 3D points
5. Select physically valid pose where:

$$z_1 > 0$$

$$z_2 > 0$$

for both cameras